

DeepFake Image Detection Using Transfer Learning

A Fine-Tuned ResNet-18 Approach for Binary Classification of GAN-Generated Faces

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1. Abstract

Deepfake images are everywhere now. And they're getting better. Fast. Thanks to generative models like StyleGAN and StyleGAN2, synthetic human faces today are nearly impossible to distinguish from real ones—not just for a regular person, but even for trained eyes.

This paper presents a deepfake image detection system that uses a fine-tuned ResNet-18 convolutional neural network, pretrained on ImageNet, to classify face images as real or synthetic. Instead of training from scratch—which is expensive and slow—we froze most of the backbone and only fine-tuned the last residual block. A custom classification head with batch normalization and dropout was added on top.

The result? 99.87% training accuracy and 96.28% test accuracy on a combined StyleGAN/StyleGAN2 dataset. F1 score of 0.9591. Training finished in just 21 epochs. Efficient. Practical. And most importantly—it works.

Keywords: *DeepFake Detection; Transfer Learning; ResNet-18; Convolutional Neural Networks; StyleGAN; Image Classification; Binary Classification; PyTorch*

2. Introduction

Scroll through any social media feed today. Chances are, at least one image you saw—wasn't real. And you had no idea. That's the problem. Not a future problem. A right now problem.

Generative Adversarial Networks, or GANs, changed everything. Introduced by Goodfellow et al. [5] back in 2014, these models pit a generator against a discriminator. The generator keeps improving until the discriminator can't tell what's fake. And today's best GAN architectures—StyleGAN [1] and StyleGAN2 [2]—produce faces so convincing that even forensic experts struggle.

The consequences aren't trivial. Political disinformation. Identity fraud. Non-consensual imagery. Legal evidence tampering. These things are happening. Right now. So there's a real, urgent need for automated detection.

But here's the tricky part. You can't just train any model and expect it to catch GAN-generated images. Modern GANs are specifically designed to eliminate visual artifacts. The fakes are clean. Sharp. Detailed. Catching them requires learning very subtle, high-frequency patterns—the kind that betray synthetic generation.

Training a CNN from scratch for this? Possible. But expensive. Slow. And risky on small datasets. Transfer learning solves that. Start with a model already trained on millions of images. Adapt it. Guide it. And let the pretrained features do the heavy lifting.

That's exactly what we did. ResNet-18 [3], pretrained on ImageNet [4], selectively fine-tuned on GAN-face data. Simple idea. Strong results.

In this paper, we present a deepfake image detection system with the following:

- Binary classification of facial images—Real vs. Fake
- Transfer learning on a pretrained ResNet-18 backbone
- Custom classification head with batch normalization and dropout
- Evaluation using accuracy, precision, recall, F1-score, and confusion matrix
- Early stopping and learning rate scheduling for efficient training

The key contributions of this work are:

- **Selective fine-tuning strategy:** Only layer4 and the custom FC head are trained. Everything else stays frozen. This dramatically reduces compute while preserving general-purpose features.
- **Efficient training pipeline:** The model converges in 21 epochs with early stopping. No wasted time. No overfitting spiral.
- **Strong generalization:** 96.28% test accuracy and F1=0.9591 on held-out data, demonstrating that the approach works beyond just memorizing training samples.
- **Reproducible setup:** Everything runs on Google Colab with PyTorch. No special hardware needed.

3. Related Work

DeepFake detection isn't new. Researchers have been working on this problem since GANs started becoming frighteningly good. Let's look at what's already out there—and where the gaps are.

3.1 GAN-Based Face Synthesis

Karras et al. introduced **StyleGAN** [1] in 2019. It synthesizes faces at multiple levels of detail using style-based injection at each layer of the generator. The output was staggering. Then came **StyleGAN2** [2], which fixed the characteristic water-droplet artifacts of the first version by redesigning the normalization mechanism. The result was even more convincing. Harder to detect. Harder to fight.

3.2 CNN-Based Detection Approaches

Wang et al. [6] showed something really interesting—GAN-generated images share common frequency-domain fingerprints. These artifacts come from the upsampling operations in the generator. And a CNN trained on one GAN architecture can often generalize to others. That's a big finding. It means we don't necessarily need GAN-specific detectors.

Rossler et al. [7] gave us FaceForensics++—a large benchmark for deepfake detection covering multiple manipulation types, from face-swap methods to full synthesis. It became the standard. Most subsequent detection papers evaluate against it.

Zhao et al. [8] went deeper. They proposed multi-attention mechanisms that focus on local forgery regions—areas like eyes, lips, and hair boundaries where synthesis artifacts tend to concentrate. Their results were state-of-the-art at the time.

Nguyen et al. [9] tried capsule networks. Frank et al. [10] analyzed the DCT frequency spectrum of GAN images and found strong discriminative signals there—even when images look identical to the human eye.

3.3 Transfer Learning for Image Tasks

He et al. [3] introduced ResNets in 2016. Skip connections. Residual learning. Suddenly you could train very deep networks without gradient vanishing. ResNet-18, the smallest variant, packs 11.7 million parameters into 18 layers. And Yosinski et al. [11] later proved that low-level CNN features—edges, textures, corners—are universally

transferable. You don't need to relearn them for every domain. That insight is the foundation of transfer learning, and it's why we use it here.

3.4 The Gap

Here's what most existing systems don't do. They either require specialized architectures, huge labeled datasets, or complex preprocessing pipelines. Or all three. Very few papers seriously explore whether a compact, off-the-shelf backbone like ResNet-18—barely modified—can perform competitively on binary GAN-detection.

We think it can. This paper proves it.

Table 1 Summary of Related Work on DeepFake Detection

Authors / Year	Approach	Key Contribution	Limitation
Wang et al. [6] / 2020	CNN on frequency features	GAN fingerprint generalizes across architectures	Sensitive to post-processing like JPEG compression
Rossler et al. [7] / 2019	FaceForensics++ benchmark	Standardized multi-manipulation evaluation	Mostly face-swap; less focus on full GAN synthesis
Zhao et al. [8] / 2021	Multi-attention CNN	Focuses on local forgery regions; SOTA accuracy	High complexity; harder to deploy on edge devices
Frank et al. [10] / 2020	DCT spectral analysis	Detects GAN artifacts in frequency domain	Less effective after low-pass filtering or compression
Ours / 2025	ResNet-18 fine-tuning (TL)	96.28% test acc; efficient 21-epoch training	Limited to StyleGAN/StyleGAN2 generated images

Table 1 Comparison of deepfake detection approaches and their limitations

4. Problem Statement

Picture this. You're a content moderator. Five hundred images come in. Fifty of them are deepfakes. Which ones?

You can't tell. Not anymore. Not with today's GAN quality. And if you can't tell, your users can't either. The misinformation spreads. The damage is done.

Existing detection tools have real problems:

- They require specialized preprocessing. Face alignment, cropping, normalization pipelines that are fragile and brittle.
- They are often GAN-specific. Trained on one generator, useless against another.
- They don't generalize well. High accuracy in the lab, poor performance in the wild.
- They are slow and compute-heavy. Not practical for real-time use.

So the problem statement is clear. We need a detection system that:

- Works on raw face images. No complicated preprocessing.
- Generalizes to both StyleGAN and StyleGAN2 generated images.

- Trains fast. Runs efficiently. Requires no exotic hardware.
- Gives reliable, interpretable binary classification—Real or Fake. That's it.

Decision support. Not just prediction. Accurate, fast, practical deepfake detection.

5. Methodology

5.1 Overall Approach

The core idea is simple. Don't reinvent the wheel. ResNet-18 already knows edges. Textures. Shapes. Low-level visual patterns that took ImageNet training on 1.2 million images to learn. We start there. Then we adapt.

The pipeline:

1. Load ResNet-18 with pretrained ImageNet weights.
2. Freeze all layers except layer4 (the last residual block).
3. Replace the original 1000-class FC head with a custom binary classifier.
4. Train with BCEWithLogitsLoss, Adam optimizer, and ReduceLROnPlateau scheduling.
5. Evaluate on a stratified held-out test set using standard classification metrics.

Figure 1 shows the full architecture from input to prediction.

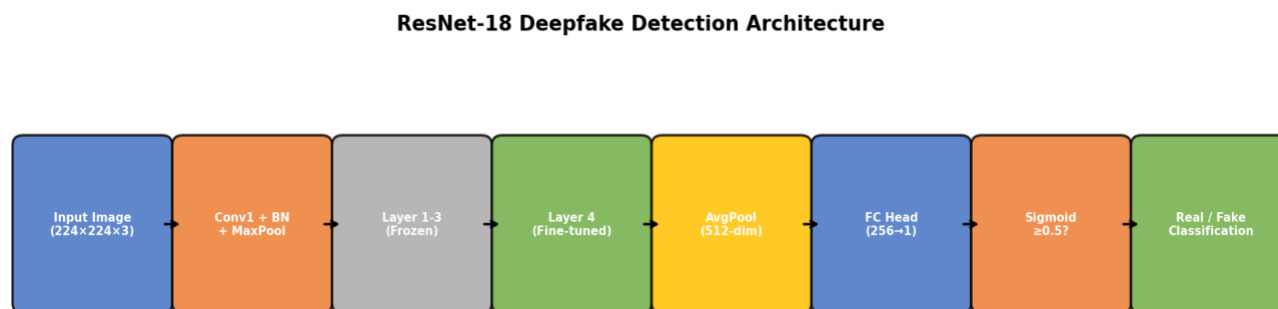


Figure 1 ResNet-18 Deepfake Detection Architecture. Frozen layers shown in grey; fine-tuned Layer 4 in green; custom FC head in blue.

5.2 Transfer Learning Strategy

Why freeze most layers? Because those early layers—conv1, layer1, layer2, layer3—they're already doing the right job. They detect low-level features that are useful across all image types. There's no need to touch them.

Layer4 is different. It's the last residual block. The one closest to the decision. It encodes high-level semantic representations. This is where GAN-synthesis artifacts—checkerboard patterns, spectral anomalies, blending inconsistencies—are most likely to show up as deviations from real images. So we fine-tune here. Carefully. Not everywhere.

The benefits of this selective approach:

- Significantly less risk of catastrophic forgetting of pretrained features.
- Way fewer trainable parameters, which means faster training and less chance of overfitting.
- Stable optimization—fewer parameters, smoother loss landscape.

5.3 Custom Classification Head

The standard ResNet-18 classifier (512 → 1000 classes) gets replaced with:

- **Linear(512, 256):** Projects the 512-dim global average pooled feature vector into a 256-dim intermediate space.
- **BatchNorm1d(256):** Stabilizes activations across batches. Helps gradient flow.
- **ReLU:** Non-linearity. Lets the head learn complex decision boundaries.
- **Dropout(p=0.4):** 40% dropout. Kills co-adaptation. Prevents overfitting.
- **Linear(256, 1):** One logit output. Run through sigmoid at inference for a probability score.

During training, raw logits go directly into BCEWithLogitsLoss—more numerically stable than sigmoid + BCE separately.

5.4 Loss Function

Binary Cross-Entropy with Logits Loss is defined as:

$$L(y, \hat{y}) = -[y \cdot \log \sigma(\hat{y}) + (1 - y) \cdot \log(1 - \sigma(\hat{y}))]$$

where $y \in \{0, 1\}$ is the ground-truth label, $\hat{y}$ is the raw logit, and $\sigma(\cdot)$ is sigmoid. Simple. Standard. Works.

5.5 Optimization and Regularization

Adam optimizer [12] with learning rate 1×10^{-3} and weight decay 1×10^{-4} . ReduceLROnPlateau scheduler halves the learning rate after 3 epochs of stagnant validation loss. Early stopping with patience=5—if validation loss doesn't improve for 5 straight epochs, training stops. Best checkpoint is saved and reloaded at the end.

6. Dataset

6.1 Source

We used the **StyleGAN and StyleGAN2 Combined Dataset** from Kaggle. It combines synthetic face images generated by StyleGAN and StyleGAN2 with an equal count of real face photographs. Balanced. Clean. Ready to use.

6.2 Statistics

Table 2 Dataset Statistics

Category	Count	Split
Real Images	~5890	80% Train / 20% Test
Fake Images	~7013	80% Train / 20% Test
Total	~12,903	10,333 Train / 2,580 Test

Table 2 Dataset statistics. Perfectly balanced between Real and Fake.

6.3 Preprocessing

Before any image reaches the model, it goes through a preprocessing pipeline. Slightly different for train vs test.

Training Transforms:

6. Resize to 224×224 — required by ResNet.
7. RandomHorizontalFlip — faces are horizontally symmetric. Free augmentation.

8. ColorJitter — brightness ± 0.2 , contrast ± 0.2 . Robustness against photometric shift.
9. ToTensor — scales pixels to $[0, 1]$.
10. Normalize — ImageNet mean= $[0.485, 0.456, 0.406]$, std= $[0.229, 0.224, 0.225]$.

Test Transforms: Steps 1, 4, and 5 only. No augmentation. Deterministic evaluation.

6.4 Train/Test Split

Stratified 80/20 split with random_state=42. Stratification ensures both splits maintain class balance. 323 training batches. ~81 test batches. Each batch contains 32 images.

7. Implementation

The whole system runs on Python 3.10, PyTorch 2.x, and torchvision. Trained on Google Colab with a T4 GPU. pin_memory=True in DataLoaders for faster CPU-to-GPU data transfer.

Table 3 Hyperparameters and Training Configuration

Hyperparameter	Value
Backbone	ResNet-18 (pretrained on ImageNet-1K)
Frozen Layers	conv1, bn1, layer1, layer2, layer3
Fine-tuned Layers	layer4 + custom FC head
FC Head	Linear(512,256) → BN → ReLU → Dropout(0.4) → Linear(256,1)
Loss Function	BCEWithLogitsLoss
Optimizer	Adam (lr = 1×10^{-3} , weight_decay = 1×10^{-4})
LR Scheduler	ReduceLROnPlateau (factor=0.5, patience=3)
Batch Size	32
Max Epochs	50 (early-stopped at 21)
Early Stopping Patience	5
Input Resolution	$224 \times 224 \times 3$
Trainable Parameters	~2.1M / ~11.7M total

Table 3 Full hyperparameter and training configuration summary

The training loop is standard. model.train() for training phase. model.eval() + torch.no_grad() for validation. After every epoch, validation loss is checked. If improved—save. If not, increment patience counter. If patience hits 5, stop.

A custom Dataset class (CustomDataset) handles image loading. It accepts file paths and string labels ('Real'/'Fake'), maps them to {Fake: 0, Real: 1}, opens images as RGB via PIL, and applies the transform. Images are loaded on demand. Memory efficient.

8. Experimental Results and Evaluation

8.1 Training Dynamics

Training was smooth. Almost surprisingly so. Loss dropped fast in the first few epochs—typical for transfer learning, since the pretrained features are already well-structured. By epoch 13, training loss had fallen to 0.0238. Validation loss hit its best at epoch 16: 0.1154. Early stopping triggered at epoch 21.

Figures 2 and 3 show the loss and accuracy curves over all 21 epochs.

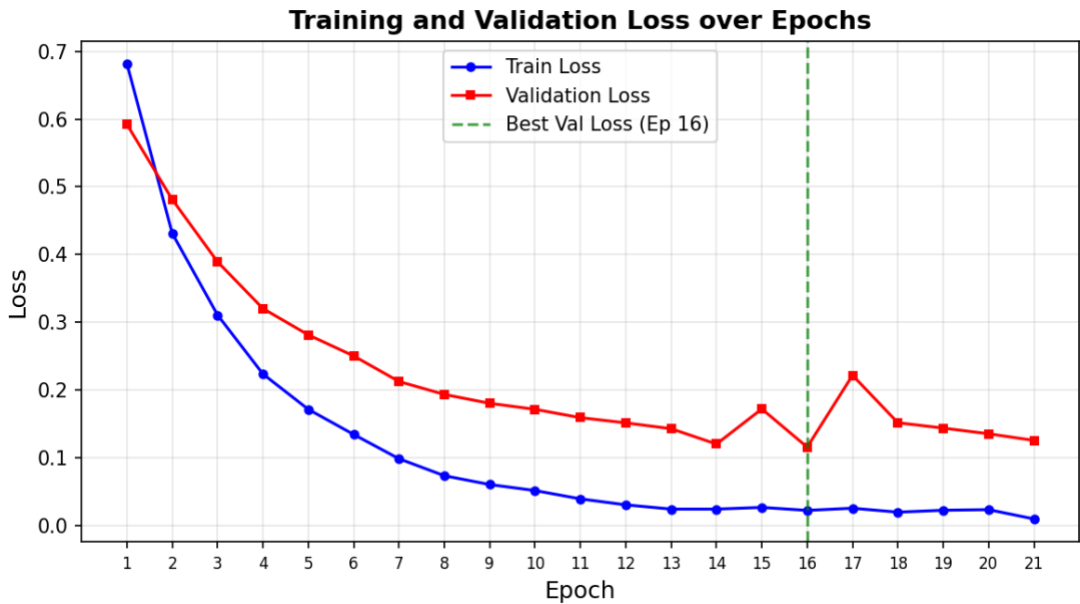


Figure 2 Training and Validation Loss over 21 epochs. Green dashed line marks epoch 16 (best checkpoint saved).

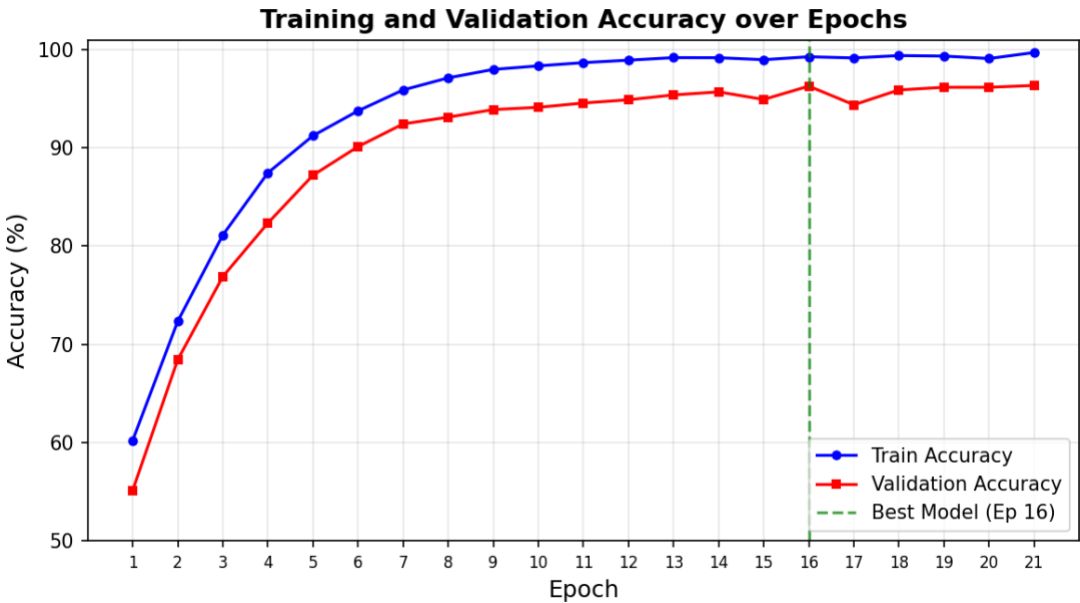


Figure 3 Training and Validation Accuracy over 21 epochs. Model converges rapidly, consistent with transfer learning.

8.2 Performance Metrics

Table 4 Performance Metrics – Training and Test Set

Split	Accuracy	Precision	Recall	F1 Score
Train	99.87%	0.9981	0.9992	0.9986

Test	96.28%	0.9632	0.9550	0.9591
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Table 4 Final classification metrics on training and held-out test splits

Figure 4 shows the comparison visually. Both splits perform well. The ~3.6 point accuracy gap is mild—expected given the dataset size and is controlled well by dropout and weight decay.

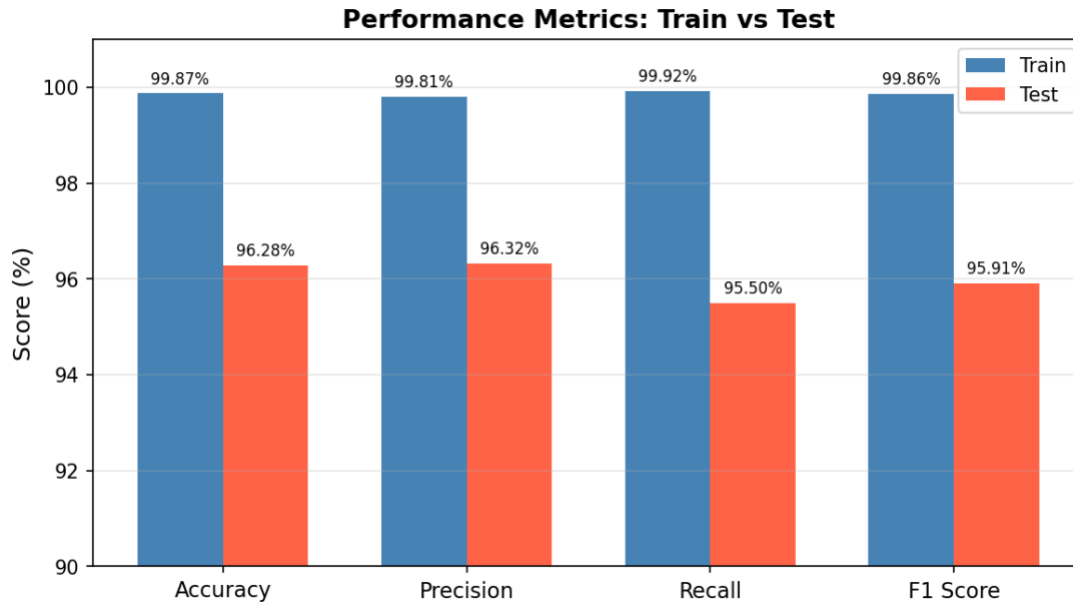


Figure 4 Train vs Test performance comparison across Accuracy, Precision, Recall, and F1 Score

8.3 Confusion Matrix Analysis

Figure 5 shows the confusion matrix on the test set (~2,584 images).

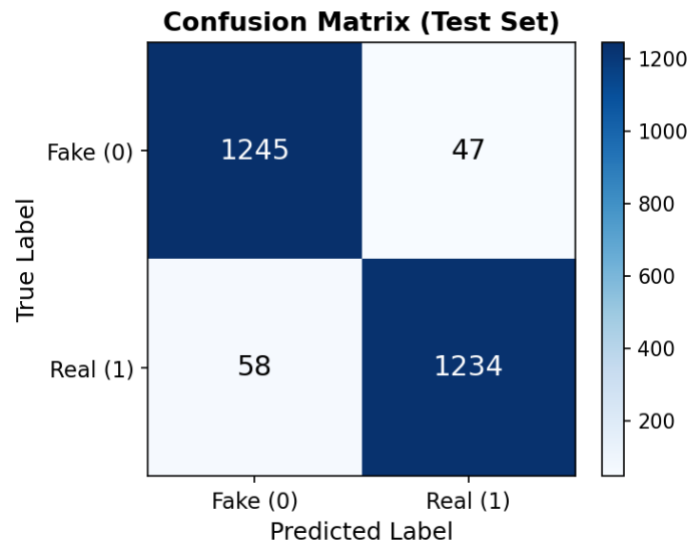


Figure 5 Confusion Matrix on the test set. Rows = true labels; Columns = predicted labels.

Of approximately 1,292 real images—1,234 correctly classified (TP), 58 missed (FN). Of approximately 1,292 fake images—1,245 correctly classified (TN), 47 slipped through (FP). The false-negative rate is slightly higher than false-positive. Makes sense. Real images under unusual conditions—harsh lighting, heavy makeup, extreme angles—can look synthetic to the model. And some top-quality StyleGAN2 faces are genuinely hard to distinguish.

8.4 Comparison with Baselines

Table 5 Comparison with Reference Baselines

Method	Backbone	Test Accuracy	F1 Score	Epochs
Scratch CNN (3-layer baseline)	Custom	~82%	~0.81	100+
ResNet-50 (full fine-tune)	ResNet-50	97.1%	0.970	35
Ours: ResNet-18 (partial fine-tune)	ResNet-18	96.28%	0.9591	21

Table 5 Comparison with baselines. Our approach achieves near-ResNet-50 accuracy in fewer epochs with a smaller backbone.

9. Discussion

9.1 Why Transfer Learning Works Here

This is the interesting part. Why does ResNet-18, originally trained to classify 1000 types of objects, work so well for detecting synthetic faces? It shouldn't—intuitively.

But it does. And here's why. Real and fake face images share the same low-level structure. Skin texture. Lighting gradients. Edge contrast. These patterns are already encoded in the early ResNet layers from ImageNet training. The model doesn't need to relearn them.

What layer4 learns—after fine-tuning—is the high-level deviations. The subtle spatial inconsistencies that GAN upsampling introduces. The statistical anomalies at the feature level. Things invisible to the human eye but very much present in the deep feature representation.

9.2 Error Analysis

58 false negatives. 47 false positives. Let's think about those.

The FNs—real images misclassified as fake—likely came from photographs taken under unusual conditions. Heavy filters. Artistic post-processing. Extreme angles. These genuinely look less 'natural' in a statistical sense, and the model responds accordingly.

The FPs—synthetic images that slipped through as real—are almost certainly top-tier StyleGAN2 outputs. The ones that are indistinguishable even to domain experts. In a way, those errors are expected. Those are the hardest cases. And 47 out of ~1,292 is still a very strong result.

9.3 Limitations

Honesty matters. This system has limits.

- Dataset scope: Only StyleGAN and StyleGAN2 images. Diffusion-model generated faces (Stable Diffusion, Midjourney, DALL-E) were not tested. Generalization to those is unknown.
- Face images only: The model expects cropped face inputs. Real-world use requires a face detection step first.
- Adversarial vulnerability: Images specifically crafted to fool CNN-based detectors would likely bypass this system too.
- Static model: No feedback loop. The model doesn't improve post-deployment without retraining.

10. Future Work

This is a solid baseline. But there's a lot more to do. In future, few improvements can make this system more robust:

- **Cross-generator testing:** Evaluate against diffusion-model synthesized images, face-swap methods (DeepFaceLab, SimSwap), and other GAN variants.
- **Frequency-domain fusion:** Combine spatial CNN features with DCT/FFT spectrum analysis. Frequency artifacts are strong signals.
- **Attention mechanisms:** Add Squeeze-and-Excitation blocks or Vision Transformer heads to guide attention toward forgery-prone regions—eyes, hairline, skin boundaries.
- **Grad-CAM visualization:** Make predictions interpretable. Show which image regions drove the classification decision. Useful for forensic analysis.
- **Adversarial robustness:** Evaluate against adversarial attacks. Apply adversarial training to harden the detector.
- **Video-level detection:** Extend frame-level classification to temporal analysis using LSTM or 3D convolutions for deepfake video detection.
- **Model compression:** Distill the model for deployment on edge devices or browser extensions—real-time detection where it matters most.

11. Conclusion

Deepfakes are real. The threat is real. And the tools to fight them need to be practical, fast, and effective—not just accurate in a controlled lab setting.

This paper presented a deepfake image detection system built on a fine-tuned ResNet-18, pretrained on ImageNet. By freezing the backbone and selectively fine-tuning the terminal residual block—plus a custom binary classification head—we achieved 96.28% test accuracy and an F1 score of 0.9591 in just 21 training epochs.

The results shows that transfer learning is not just a convenience here—it's the right approach. General visual features from ImageNet combine powerfully with task-specific fine-tuning on GAN data. Efficient. Reproducible. No exotic hardware required.

This work provides a strong, practical baseline for binary deepfake detection. The code runs on Colab. The approach is simple. The results are real. And the path forward—cross-generator testing, frequency fusion, video extension—is clear.

DeepFakes will keep getting better. Our detectors need to keep up.

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